MGT 3213

Management Information Systems

■ 8 Modules | ■ Dr. A. Rukshan | ■■ University of Vavuniya | ■ Complete Study Notes

#	Module Title
01	Basics of Management Information Systems
02	The Strategic Role of Information Systems
03	Ethics & IS Strategy
04	Information Systems Planning (ISP)
05	Information Systems Development (ISD)
06	Database Approach to Data Management
07	Data Communications & Networking
08	Cloud Services for MIS

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MOD 01 BASICS OF MANAGEMENT INFORMATION SYSTEMS

What is an Information System (IS)?

An IS is any organized combination of **people, hardware, software, communications networks, data resources, and policies/procedures** that collects, organizes, stores, retrieves, transforms, and disseminates information in an organization.

The DIKW Pyramid

A hierarchy showing how raw data becomes actionable wisdom:

Level	Description	Understanding
Data	Raw, unprocessed facts	None
Information	Data with context; relationships understood	Relations
Knowledge	Patterns identified from information	Patterns
Wisdom	Principles understood; highest-value output	Principles

■ ***Key Insight:*** *Each step up the DIKW pyramid adds understanding and value to initial data. Moving upward also increases 'connectedness' — the ability to apply knowledge across broader contexts.*

Data vs. Information

Concept	Description	Example
Data	Raw, unprocessed facts	List of names and numbers
Information	Data organized meaningfully	Table with Employee_Name & ID cc

Types of Information by Organizational Level

Level	Type	Decision Structure	IS Used
Strategic (Top Mgmt)	Unstructured	Unstructured	Decision Support System
Tactical (Middle Mgmt)	Semi-structured	Semi-structured	Management Information Sys
Operational (Lower Mgmt)	Structured	Structured	Transaction Processing Syst

IS Sophistication Levels

Level	Technology	Best For
Level 1	Paper-based system	Very small, simple operations
Level 2	Desktop apps (Word, Excel)	Small information volume
Level 3	Databases & analysis packages	Medium information volume

Level	Technology	Best For
Level 4	Fully integrated systems (ERP etc.)	Large, complex organizations

> ■ ***Key Insight:*** *Each level drops in efficiency as information volume grows. Managers must identify when to upgrade to the next IS level.*

Three Major Roles of IS in Organizations

- **Support business processes and operations** — automates routine transactions
- **Support business decision-making** — provides analytical reports
- **Support competitive advantage** — enables strategic differentiation

Types of Information Systems

Abbreviation	Full Name	Primary Purpose
TPS	Transaction Processing System	Daily routine transactions
OAS	Office Automation System	Productivity & communication
MIS	Management Information System	Managerial reporting
EIS	Executive Information System	Top-level strategic dashboards
DSS	Decision Support System	What-if analysis & decisions
ES	Expert System	AI-based specialist knowledge
WGSS	Work Group Support System	Team collaboration
FIS	Financial Information System	Financial planning & reporting
PIS	Project Management IS	Project tracking & control

Definition & Functions of MIS

MIS = an organized system for **collecting, storing, processing, and delivering information** to support management operations and decision-making.

Functions of MIS

Function	Description
Data Collection	Gathering data from various sources (sales, HR, finance)
Data Processing	Converting raw data into meaningful information
Information Storage	Keeping data in databases for future retrieval
Information Dissemination	Distributing reports and summaries to managers
Decision Support	Helping managers make informed decisions via analytics

MIS Sub-Systems

Sub-System	Tracks / Manages
Sales MIS	Sales performance by region, product, or salesperson
Financial MIS	Budget reports, profit analysis, financial statements
HR MIS	Employee records, payroll, performance data
Stock/Inventory MIS	Stock levels, reorder points, supplier performance

Benefits of MIS

- Improves **decision-making quality**
- Increases **efficiency and productivity**
- Enhances **communication and coordination**
- Provides **real-time and accurate information**
- Helps in **strategic planning and forecasting**

MOD 02 THE STRATEGIC ROLE OF INFORMATION SYSTEMS

Key Definitions

Term	Definition
Strategy	Making choices and creating a unique position by performing different activities fr
Strategic Role	How an activity, resource, or system contributes to achieving long-term goals, sus
Competitive Advantage	Ability to perform better than competitors by offering greater value through lower p

Why IS Matters Strategically

- Improve efficiency & productivity
- Enable better decision-making
- Create competitive advantages
- Transform business models (e.g., Uber, Airbnb)
- Support globalisation

Evolution of IS — From Back-Office to Boardroom

Era	Role & Focus
1960s–70s	Data Processing — Transaction automation, cost reduction
1980s	Management Reporting — MIS, DSS for managerial control
1990s	Strategic Weapon — SIS, ERP, the Internet
2000s	Collaborative Core — Integration, supply chains, CRM
2010s–Now	Digital Foundation — Platforms, analytics, AI, ecosystem driver

Three Levels of IS Role

Role	Focus	Tools / Examples
Operational	Routine daily tasks	Payroll systems, inventory updates
Managerial	Monitoring & controlling performance	MIS dashboards, reports
Strategic	Long-term planning & competitive edge	ERP, AI, Big Data analytics

■ ***Key Insight:*** *Example: Sri Lankan supermarkets use POS (operational) + ERP (strategic) systems together for demand forecasting and inventory management.*

Porter's Five Forces & IS

IS can **reduce competitive pressures** across all five forces:

Force	How IS Helps	Real Example
Buyer Power	Loyalty apps, personalisation reduce switching	Amazon Prime
Supplier Power	SCM systems give firms more alternatives	Walmart's Retail Link
New Entrants	AI-driven ops create high entry barriers	Uber's algorithm
Substitutes	Data-driven differentiation reduces substitution	Netflix recommendations
Rivalry	Cost leadership through information efficiency	Toyota manufacturing

Porter's Value Chain & IS

Activity	IS Application	Example
Inbound Logistics	RFID, IoT tracking	Real-time shipment visibility
Operations	Automation, robotics	Smart manufacturing systems
Outbound Logistics	Route optimisation	UPS ORION routing algorithm
Marketing & Sales	CRM, digital advertising	Salesforce CRM
Service	Chatbots, online support	AI-powered help desks

■ ***Key Insight:*** *ZARA's success is rooted in real-time inventory management IS, not just fashion trends. Their system gets a garment from design to store shelf in 25 days.*

Resource-Based View (RBV) & IS — VRIO Framework

Criterion	Question	Outcome if YES
Valuable (V)	Does IS help exploit opportunities or neutralise threats?	Competitive parity or advantage
Rare (R)	Is the IS rare among current/potential competitors?	Temporary competitive advantage
Inimitable (I)	Is the IS difficult/costly to imitate?	Sustained competitive advantage
Organised (O)	Is the firm organised to exploit the IS capability?	Full realisation of advantage

■ ***Key Insight:*** *Netflix's proprietary viewing-data recommendation algorithm scores high on all four VRIO dimensions, providing a truly sustained competitive advantage.*

Strategic Information Systems (SIS)

A **Strategic Information System (SIS)** is an information system developed or used to **support or shape the competitive strategy** of an organisation.

Regular IS	Strategic IS (SIS)
Supports routine operations	Changes the way the organisation competes
Improves efficiency	Creates competitive advantage
Short-term focus	Long-term focus
Back-office focus	Customer, supplier & competitive focus

Regular IS	Strategic IS (SIS)
Automates existing processes	Innovates and disrupts
Payroll, attendance, inventory	Uber surge algo, Zara real-time IS, Walmart RFID

MOD 03 ETHICS & IS STRATEGY

Ethics vs. Strategy

Dimension	Strategy	Ethics
Focus	What a firm does to succeed	How a firm achieves its goals
Driver	Profit, market share, growth	Moral values, fairness, responsibility
Time Horizon	Short to long-term gains	Long-term societal & reputational impact

■ ***Key Insight:*** *A purely profit-driven strategy without ethics can lead to short-term gains but long-term reputational and legal risks — this is called Strategic Myopia.*

Ethical Strategy

An ethical strategy ensures that **how goals are achieved matters as much as the goals themselves**. It aligns competitive advantage with:

- **Moral values** — acting with integrity and respect
- **Legal compliance** — following laws and regulations
- **Social and environmental responsibility** — considering all stakeholders

Key Ethical Challenges in IS

Challenge	Description
Data Privacy & Confidentiality	Organisations collect large volumes of user data for analytics and monetis
Surveillance & Employee Monitoring	Excessive monitoring through IS can violate employee privacy and autono
Customer Manipulation	Using behavioural data and algorithms to exploit or manipulate customer
Misuse of IS	Using systems to escape regulations or social responsibilities

Real-World Ethical Case Studies

Case	Year	Ethical Issue	Consequence
VW Emissions Scandal	2015	Software used to cheat emission tests (escape regulations)	Billions in fines, repu
Facebook–Cambridge Analytica	2018	User data misused for political manipulation without consent	Global regulatory bac
Uber 'Greyball' Program	2017	Fake app used to evade law enforcement across Europe	DOJ investigation, fo

Principles of Ethical Strategy

Principle	Description
Transparency	Clear, honest communication with all stakeholders
Fairness	Equitable treatment of employees, customers, and suppliers

Principle	Description
Accountability	Accepting full responsibility for business outcomes and impacts
Sustainability	Considering long-term social, environmental, and economic impact

MOD 04 INFORMATION SYSTEMS PLANNING (ISP)

What is ISP?

ISP is a **strategic, structured process** that aligns an organization's IT infrastructure and applications with its overall business goals, goals, and strategies. It involves developing a roadmap to optimize information, technology, and resources for competitive advantage.

Three Key Activities of ISP

Activity	Description
1. Describe Current Situation	Model existing locations, units, functions, processes, data, and IS
2. Describe Target Situation	Define desired future state — focus on differences from current lis
3. Develop Transition Plan	Short-term & long-term plans to move from current to desired state

Planning Approaches

Approach	Description	Focus
Top-Down Planning	Broad understanding of the entire organization's IS needs	Whole enterprise view
Bottom-Up Planning	Identifies IS projects based on solving specific operational problems	Individual problem-solving

IS Planning Process (Flow)

Step	Process Element
1	Organization Mission + Business Assessment
2	Organizational Strategic Plan + Current IT Architecture
3	IS Strategic Plan
4	New Information Technology Architecture
5	IS Operational Plan
6	IS Development Projects

Information Collected About Current Situation

Item	Description
Locations	Where the company operates
Business Units	Organisational units that operate within the company
Functions	Cross-organisational activities for day-to-day operations
Processes	Specific processes of the company

Item	Description
Data	Data underlying all processes
Information Systems	Automated and non-automated systems supporting business processes

MOD 05 INFORMATION SYSTEMS DEVELOPMENT (ISD)

Definition

ISD is the comprehensive process of **planning, creating, implementing, and maintaining** information systems (software, databases, networks) to support an organization's strategic goals using various methodologies.

Stages in ISD

Stage	Activities	Tools / Techniques
Planning & Analysis	Define needs, objectives, feasibility, requirements	Questionnaires, interviews, use-case di
Design	Blueprints: architecture, UI, data structures	Wireframes, design modelling, ER diagr
Development	Writing code, configuring software, building system	IDEs, frameworks, version control
Testing	Verify system works correctly and meets requirements	Unit tests, integration tests, UAT
Deployment/Maintenance	Launch system and provide ongoing support/updates	CI/CD pipelines, monitoring tools

ISD Methodologies Comparison

Methodology	Approach	Best For	Key Risk
Waterfall	Sequential, linear phases	Well-defined, stable requirements	Late discovery of defects
Agile	Iterative sprints, flexible	Changing requirements, collaboration	Scope creep, less document
V-Model	Waterfall + parallel testing phases	Safety-critical systems	Rigid like Waterfall
Spiral	Risk-driven, iterative loops	Large, complex, risky projects	Complex to manage, costly

Waterfall Model — Detailed

Sequential phases: **Requirements** → **System Design** → **Development** → **Testing Cycles (UT/IT/ST/UAT)** → **Deployment** → **Maintenance**

Cons of Waterfall

- Inflexibility — hard to accommodate changes mid-project
- Late Testing and Integration
- Poor Adaptability to Change
- Delayed Delivery — no working product until late stages
- Risk of Misalignment with User Needs

Waterfall Variations

Variation	Key Feature
Modified Waterfall	Adds feedback loops — allows minor revisions to previous phases
Sashimi Model	Allows slight phase overlaps for more fluid transitions

Variation	Key Feature
Incremental Waterfall	Divides project into smaller segments, each delivered incrementally
Agile-Waterfall Hybrid	Planning/requirements = Waterfall; Development/testing = Agile sprints

Spiral Model — Four Phases Per Cycle

Phase	Activities
1. Planning	Determine objectives, alternatives, and constraints for the cycle
2. Risk Analysis	Evaluate alternatives, identify and resolve risks, create prototypes
3. Engineering	Develop and verify the next-level product (design, code, test)
4. Evaluation	Customer evaluates current results and plans the next iteration

Advantages of Spiral Model

- Best model to handle **risky projects**
- Good choice for **large projects**
- Flexible with requirements
- Better customer satisfaction through iterative feedback
- Open to receiving feedback at every cycle
- Better quality through continuous testing

MOD 06 DATABASE APPROACH TO DATA MANAGEMENT

Flat File vs. Relational Database

Feature	Flat File	Relational Database
Structure	Single table	Multiple linked tables
Redundancy	High — data repeated	Low — data stored once
Update	Must update every row	Update one record, reflects everywhere
Example	Boss/phone repeated per employee	Separate STAFF & DEPARTMENTS tables

RDBMS Key Terminology

Term	Meaning
Column / Field / Attribute	A property/characteristic of an entity (e.g., Employee_Name)
Row / Record / Tuple	A single entry in a table (e.g., one employee's data)
Primary Key	A unique identifier for each record
Domain	The range of valid values for an attribute (e.g., 5000–100000 for salary
Degree	The number of columns in a table
Cardinality	The number of rows (records) in a table

DBMS (Database Management System)

Software that acts as an **interface between application programs and data files**.

Feature	Description
Interface	Acts as bridge between application programs and data files
Logical/Physical Separation	Separates how data is perceived (logical) from how it is stored (physical)
Multiple Views	Allows different logical views for different user groups
Redundancy Control	Reduces data redundancy and eliminates inconsistency

Managing Data — Key Concepts

Concept	Description
Data Modelling	Designing relationships between data through ER diagrams and models
Data Mining	Extracting valuable patterns and information from large datasets
Data Integration	Combining data from multiple different sources for unified analysis
Data Governance	Policies for data handling, consistency, and compliance

Analytics Types

Type	Question Answered	Value Level	Example
Descriptive	What happened?	Hindsight	Sales reports, dashboards
Predictive	What will happen?	Insight	Demand forecasting
Prescriptive	How do we make it happen?	Foresight	Automated reordering systems

■ ***Key Insight:*** *Moving from Descriptive → Predictive → Prescriptive analytics increases both complexity and business value.*

Data Mining Models

Category	Sub-Types
Descriptive (What happened?)	Clustering, Summarization, Sequence Discovery, Association Rules
Predictive (What will happen?)	Classification, Regression, Prediction, Time Series Analysis

MOD 07 DATA COMMUNICATIONS & NETWORKING

Data Communication

Process of **exchanging data between two devices** over a transmission medium using a communication system made of hardware and software.

Five Components of a Data Communication System

Component	Description	Examples
Message	Data being communicated	Text, audio, video files
Sender	Device that sends the data	Computer, mobile phone, laptop
Receiver	Destination device	Computer, telephone, workstation
Transmission Medium	Physical path connecting sender and receiver	Cables, radio waves, fibre
Protocol	Rules governing data communication	TCP/IP, HTTP, FTP, DNS

Computer Network Types (by Size)

Type	Full Name	Coverage	Example
PAN	Personal Area Network	~10 meters (around a person)	Bluetooth devices
LAN	Local Area Network	Room/Building/Campus (10m–1km)	Office network
MAN	Metropolitan Area Network	City (~10 km)	City-wide Wi-Fi
WAN	Wide Area Network	Country/Continent (100–1000+ km)	The Internet

LAN Topologies

Topology	Description	Advantage	Disadvantage
Bus	All devices on one cable	Easy, cheap to install	If main cable fails, whole net
Ring	Devices connected in a loop	Fast; no data collisions	One device failure = entire n
Star	All connect to central hub or switch	Reliable; one cable fail = OK	Expensive; hub/switch failur

Transmission Media

Guided (Wired) Media

Medium	Description	Advantages	Disadvantages
Twisted Pair	Pairs of wires twisted together	Cheap, easy to install	Limited bandwidth, susce
Coaxial Cable	Central conductor + metallic shield	Better shielding, more bandwidth	More expensive than twi
Optical Fibre	Light through glass/plastic core	High bandwidth, lightweight, immune to EMI interference	Fragile, expensive to ins

Unguided (Wireless) Media

Medium	Frequency Range	Uses	Key Characteristic
Radiowaves	3 KHz – 1 GHz	AM/FM radio, cordless phones	Easy to generate; penetrates buildin
Microwaves	1 GHz – 300 GHz	Mobile phones, satellite, LAN	Line-of-sight; antennas must be alig
Infrared	300 GHz – 400 THz	TV remotes, wireless mouse/keyboard	Very short range; cannot penetrate
Satellite	GHz bands	Global communication, GPS	Orbits at 36,000 km; uplink/downlink

Key Protocols

Protocol	Full Name	Purpose
TCP/IP	Transmission Control Protocol / Internet Protocol	Core Internet communication protocol
HTTP	Hypertext Transfer Protocol	Web page transfer
FTP	File Transfer Protocol	File transfer between systems
DNS	Domain Name System	Translates domain names to IP addresses
DHCP	Dynamic Host Configuration Protocol	Assigns IP addresses automatically
SMTP	Simple Mail Transfer Protocol	Sending emails
POP	Post Office Protocol	Receiving emails

OSI vs. Internet Network Model

OSI Layer #	OSI Layer Name	Internet Model	Examples
7	Application	Application Layer	HTTP, FTP, web browsers
6	Presentation	Application Layer	Encryption, compression
5	Session	Application Layer	Session management
4	Transport	Transport Layer	TCP/IP software
3	Network	Network Layer	IP routing
2	Data Link	Data Link Layer	Ethernet ports, cables
1	Physical	Physical Layer	Ethernet software drivers

How Computer Networks Are Used in MIS

Use	Description
Business Process Integration	Networks integrate various business processes, reducing redundancy
Data Sharing & Collaboration	All departments access the same data for coordinated decision-making
Centralised Data Storage	Secure central storage that's easier to manage and back up
Remote Access	Authorised personnel access MIS from anywhere
Real-Time Communication	Email, messaging, video conferencing for quick decisions

Use	Description
Information Security	Access controls and encryption protect sensitive data
ERP Systems	Networks enable ERP to integrate finance, HR, supply chain across the o
CRM Systems	Networks allow CRM to collect and analyse customer data company-wide

MOD 08 CLOUD SERVICES FOR MIS

What is Cloud Computing?

Cloud computing is the **on-demand availability of computer system resources**, especially data storage and computing power, **without direct active management by the user**. It typically uses a **pay-as-you-go** model.

Types of Cloud Deployment

Type	Who Manages IT	Key Features	Example Providers
Private Cloud	Same company; not shared	High security, full control	On-premises servers
Public Cloud	Single cloud provider	Scalable, inexpensive, location-independent	Oracle OCI, AWS, Azure
Hybrid Cloud	Mix of private + public	Security + scalability + cost-effective	AWS + own data centre
Multi-Cloud	Multiple cloud providers	Avoids vendor lock-in, flexible	AWS + Azure + Google

Cloud Service Models (IaaS / PaaS / SaaS)

Model	Full Name	What Vendor Manages	Target Users
IaaS	Infrastructure as a Service	Servers, storage, networking	Infrastructure & network
PaaS	Platform as a Service	+ OS, runtime, middleware, databases	Application developers
SaaS	Software as a Service	Everything including applications	End users

■ ***Key Insight:*** *On-premise = you manage everything. IaaS = vendor manages hardware. PaaS = vendor manages platform. SaaS = vendor manages all, you just use the app.*

Advantages of Cloud Computing

- **Cost Saving** — reduces capital expenditure on hardware
- **Flexibility** — access from anywhere, any device
- **24/7 Availability** — always-on service
- **Reliability** — built-in redundancy and failover
- **Unlimited Storage** — scalable storage on demand
- **Scalability** — scale up or down based on need
- **Easy Backup & Restore** — automated backup solutions
- **Automatic Software Updates** — vendor handles maintenance
- **Performance** — high-performance infrastructure

Disadvantages of Cloud Computing

- **Security Issues** — data stored externally raises privacy concerns
- **Management Issues** — vendor dependency and control limitations
- **Lack of Expertise** — requires skilled staff to configure and manage
- **Performance Issues** — internet dependency can cause latency
- **Migration Challenges** — moving existing systems to cloud is complex

Industry-Specific Cloud MIS Applications

Industry	Cloud MIS Application
Healthcare	Track patient records, manage appointments, analyse treatment outcomes
E-commerce	Monitor website traffic, analyse purchasing patterns, optimise pricing in real-time
Financial Services	Monitor market trends, manage risk, provide personalised financial advice

MOD ✓ KEY TAKEAWAYS — ALL MODULES

#	Takeaway
1	IS is not just a tech tool — it is a strategic resource that shapes competitive advantage
2	The DIKW hierarchy (Data→Information→Knowledge→Wisdom) underlies all data management thinking
3	Organizations must choose ISD methodologies (Waterfall vs Agile) based on project complexity and risk
4	Relational databases eliminate redundancy; DBMS enables multiple logical views of shared data
5	Networks are foundational to MIS — enabling integration, automation, and real-time decisions
6	Cloud computing shifts IS from capital expense to operational expense with added flexibility
7	Ethics must be embedded in IS strategy — misuse leads to reputational and legal damage
8	ISP aligns IT with business goals through 3 steps: Current Situation → Target → Transition Plan
9	Porter's Value Chain and Five Forces both show how IS creates and defends competitive advantage
10	Strategic IS (SIS) innovates and changes how an org competes; Regular IS only improves efficiency

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